Observing, Inferring and Predicting

Word	Meaning
Observing	What I can see, smell, taste, touch and hear.
Inference	Trying to explain what I observed.
Prediction	What I think will happen.

Situation	Observation, inference or prediction?
It will rain tomorrow.	
It is raining at the moment.	
I think the tree fell down because there was a storm last night.	
The rock has a mass of 100kg.	
I will pass the Science test in term 1.	
The bike is red.	
He is yawning because he is tired.	

Types of Observation

Observation type	Definition	Example
Qualitative observation	An observation without numbers or measurements	It is hot today
Quantitative observation	An observation with number measurements	The temperature is 42°C today

Try these

Situation	Qualitative or quantitative observation?
The bike is red	
The bus has a mass of 2,000kg	
The water is 5 °C	
The water is cold	
The road is 5km long	
The hole is deep	

Candle Observation

Observation	What you observed	Is it a qualitative or quantitative

		observation?
Colour		
How does the candle feel?		
Smell before burning		
Starting Mass	g	
Height before burning	cm	
Smell while burning		
Colour of flame		
Colour of smoke		
Mass after burning for 5 minutes	g	
Height after burning for 5 minutes	cm	